

A REPORT

ON

“Java Development Intern”

Submitted by

Amrutha Shree G R - 20241CSE0754

Under the guidance of,

Ms. Shaik Salma Begum

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTERSCIENCE AND ENGINEERING

AT



PRESIDENCY UNIVERSITY

BENGALURU

AUGUST 2026

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the report of **CSE7000 – Internship** on “**Java Development**” being submitted by “**Amrutha Shree G R**” bearing roll number “**20241CSE0754**” in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Engineering** is a bonafide work carried out under our supervision.

Ms.Shaik Salma Begum
Assistant Professor
PSCS
Presidency University

Mr.Sakthi S
Assistant Professor
PSCS
Presidency University

Dr. Balakrishna
Program Intrenship Co-ordinator
PSCS
Presidency University

Dr.Bhuvaneshwari Patil
School level coordinator
PSCS
Presidency University

Dr.Asif Mohammed H B
HoD
PSCS
Presidency University

Dr. Duraipandian N
Dean – PSCS
Presidency University

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND

ENGINEERING

DECLARATION

I hereby declare that the work, which is being presented in the report entitled “**Java Development intern**” in partial fulfillment for the award of Degree of **Bachelor of Technology in Computer Science and Engineering**, is a record of my own investigations carried under the guidance of **Ms. Shaik Salma Begum** Assistant Professor and **Mr. Sakthi S** Assistant Professor, Presidency School of Computer Science and Engineering, Presidency University, Bengaluru.

I have not submitted the matter presented in this report anywhere for the award of any other Degree.

Amrutha Shree G R(20241CSE0754)

INTERNSHIP COMPLETION CERTIFICATE



ALFIDO TECH

RIDING THE WAVES OF TECHNOLOGICAL
ADVANCEMENT



CERTIFICATE OF COMPLETION

THIS CERTIFICATE IS PRESENTED TO

Amrutha Shree GR

In recognition of his/her efforts and achievements in completing
1 Month internship as a **Java Developer**.

Conducted From 1st July 2026 to 30th July 2026



Candidate ID
BS/REG/123999



Scan to Verify Certificate Authenticity

Issued by Alfido Tech – Registered Training
Organization

ACKNOWLEDGEMENT

First of all, I indebted to the **GOD ALMIGHTY** for giving me an opportunity to excel in our efforts to complete this internship on time. I express sincere thanks to our respected Dean **Dr. Duraipandian N**, Presidency School of Computer Science and Engineering , Presidency University for getting us permission to undergo the internship.

I express heartfelt gratitude to our beloved **Dr. Asif Mohammed H B**, Head of the Department, Presidency School of Computer Science and Engineering, Presidency University, for rendering timely help in completing this internship successfully.

I am greatly indebted to my reviewers **Ms. Shaik Salma Begum Assistant Professor** and **Mr. Sakthi S Assistant Professor** , Presidency School of Computer Science and Engineering, Presidency University for their inspirational guidance, and valuable suggestions and for providing a chance to express technical capabilities in every respect for the completion of the internship work.

I would like to convey gratitude and heartfelt thanks to the Internship Coordinator **Dr. Bhuvaneshwari Patil** and Program Internship Coordinator **Dr. Balakrishna R .** I also thank my family and friends for the strong support and inspiration they have provided us in bringing out this internship.

Amrutha Shree G R(20241CSE0754)

ABSTRACT

This internship, undertaken as part of the CSE7000 Internship program, focused on strengthening core Java programming skills through hands-on, real-world application development rather than isolated exercises. The primary objective was to apply fundamental programming concepts — including conditional logic, loops, collections, exception handling, and file operations — within the context of a functional software project

As part of the internship, a Library Management System was developed, evolving from a console-based application into a fully functional web application built using Java and Spring Boot. The system supports adding books to a catalog, viewing all records, and issuing and returning books, with genres tracked separately and all data persisted to a file. Key engineering practices such as input validation, duplicate-ISBN prevention, and structured exception handling were incorporated to ensure the application behaves reliably under real-world conditions.

The internship also introduced practical exposure to software development tools and workflows used in the industry, including Maven for project/dependency management and Git/GitHub for version control. Overall, the internship provided a well-rounded, practical understanding of Java application development — bridging classroom fundamentals with real-world software engineering practices.

The internship also introduced practical exposure to software development tools and workflows used in the industry, including Git and GitHub for version control. Overall, the internship provided a well-rounded, practical understanding of Java application development — bridging classroom fundamentals with real-world software engineering practices.

Introduction to Alfido Techs

Alfido Tech is a technology-driven organization that offers internship and training programs for students and fresh graduates, with a focus on strengthening practical, industry-relevant programming skills. Rather than limiting learning to theory, the company structures its internships around hands-on tasks and real-world style projects, allowing interns to apply concepts as they are learned. Its core service areas include Software Development, Java Development, Web Development, and structured Internship Training aimed at building strong technical foundations for early-career developers.

Objectives of the Project

The primary objective of the Library Management System is to develop a computerized system that simplifies and organizes the various activities performed in a library. Traditional library management involves maintaining records manually, which can be time-consuming and may result in errors or loss of information. The proposed system helps overcome these limitations by providing a digital platform for managing books, members, and transactions.

Another important objective is to centralize library information in one system. Instead of maintaining separate registers for books, students, and transactions, all relevant information can be stored in a database. This makes it easier for the librarian to access, modify, and manage records whenever required.

The system also focuses on efficient book management. Librarians can maintain details such as book name, author, category, ISBN, quantity, and availability. The system helps identify whether a particular book is available, issued, or returned, making the process of managing the library collection more organized.

The system aims to simplify the book issuing and returning process. When a book is issued to a student or member, the transaction can be recorded in the system. Similarly, when the book is returned, its status can be updated. This reduces manual calculations and makes it easier to track borrowed books.

The Library Management System also aims to improve data accuracy and reliability. Since information is stored and updated digitally, the chances of incorrect entries, duplicate records, and missing information can be reduced. Proper database management also helps maintain consistency in library records.

Additional Objectives

- 1.To provide an organized method of maintaining library records and make information easily accessible.
- 2.To simplify the process of adding and removing books from the library collection.
- 3.To provide quick access to book information through search and filtering options.
- 4.To monitor issued and returned books and maintain proper transaction records.
- 5.To maintain member records and associate borrowing activities with the respective members.
- 6.To reduce manual effort involved in maintaining registers and searching through physical records.
- 7.To improve the speed of library operations by automating repetitive tasks.
- 8.To provide accurate information about book availability whenever required.
- 9.To maintain a proper history of library transactions for future reference.
- 10.To make the system easy to use even for users with limited technical knowledge.
- 11.To provide better data organization by storing information in structured database tables.
- 12.To improve record security by controlling access to important library information.
- 13.To minimize paperwork and physical record maintenance, contributing to a more efficient digital library environment.
- 14.To make report generation easier by allowing library information and transaction details to be retrieved whenever required.
- 15.To provide a flexible system that can be modified according to the changing requirements of the library.
- 16.To improve the overall user experience by providing a simple and responsive interface.
- 17.To support future development, such as adding fine calculation, notifications, authentication, and additional user roles.

Work Progress—Jult 1 to July 30

Week 1 (July 1 – 7): Foundations & Planning

- Reviewed Core Java fundamentals — conditionals, loops, and OOP concepts
- Studied Scanner-based input handling and console I/O
- Planned the Library Management System's structure and core features

Week 2 (July 8 – 14): Collections & Exception Handling

- Implemented the Book model class with core attributes
- Applied ArrayList, HashMap, and HashSet for catalog storage, ISBN lookup, and genre tracking
- Built custom exceptions (BookNotFoundException, BookAlreadyIssuedException, InvalidInputException)
- Wrapped risky operations in try-catch blocks for safe error handling

Week 3 (July 15 – 21): File Handling & Web Application Development

- Implemented data persistence using FileWriter and FileReader
- Transitioned the project from a console-based application to a Spring Boot web application
- Built the Controller–Service layers and connected the web forms to the underlying catalog logic

Week 4 (July 22 – 30): Testing, Git/GitHub & Finalization

- Designed and refined the web interface (add/issue/return books, view catalog and genres)
- Tested the application end-to-end, including validation and error scenarios
- Created a README documenting setup and usage
- Initialized Git, committed progress, and pushed the final project to GitHub

Results

The developed Library Management System successfully provides a computerized solution for managing the major activities of a library. The system helps in maintaining book records, member details, and issue-return transactions in an organized manner. By replacing manual record maintenance with a digital system, the overall library management process becomes easier and more efficient.

The system allows the librarian to add, update, delete, and search book records whenever required. Details such as book title, author, category, ISBN, and availability can be maintained systematically. This makes it easier to identify books and check their current status without going through physical registers.

The book issue and return functionality helps maintain proper transaction records. When a book is issued, its availability status can be updated, and when it is returned, the record can be modified accordingly. This reduces confusion and helps the librarian keep track of borrowed and available books.

The system also provides better management of student/member information. User details can be stored in a centralized database, making it easier to retrieve and manage information whenever necessary. The database-based approach also helps maintain consistency and reduces the possibility of losing important records.

Another important result is the reduction of manual work and paperwork. Common library activities that previously required maintaining registers can be performed through the web application. This saves time and improves the efficiency of library operations.

The project also demonstrates the practical use of Java, Spring Boot, HTML/CSS, and MySQL in developing a database-driven web application. The application provides a simple interface through which users can perform library-related operations efficiently.

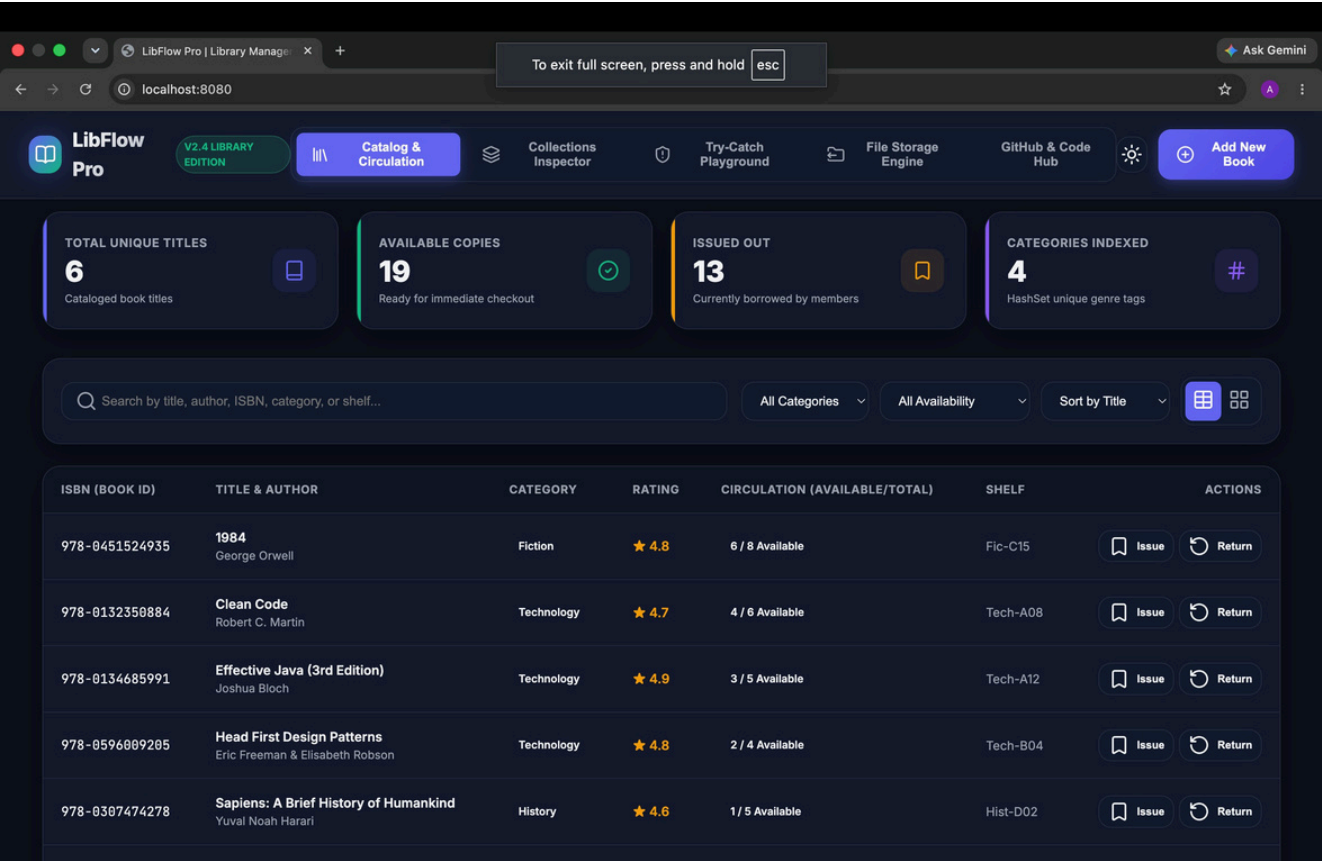


Fig 1.1

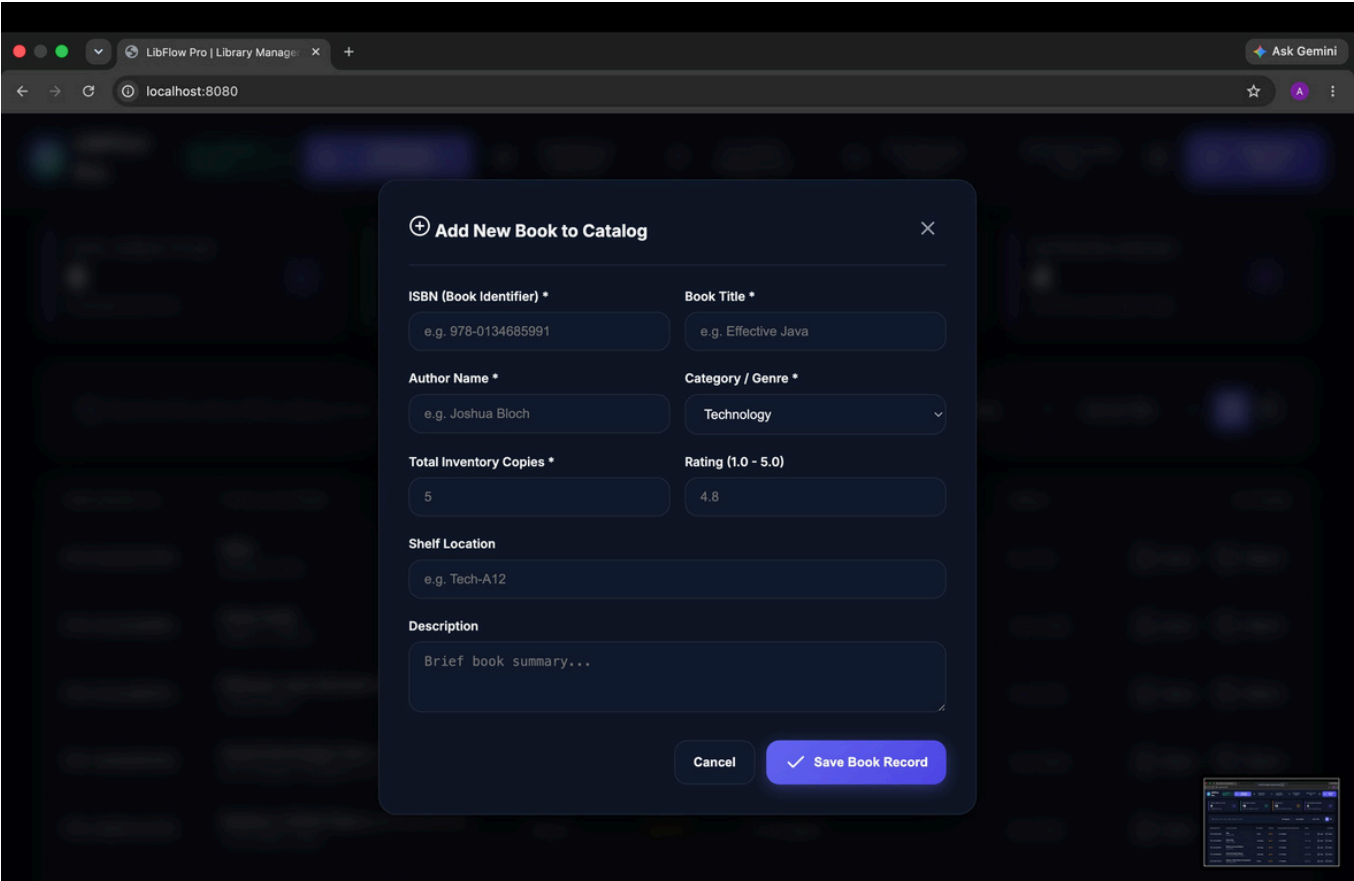


Fig 1.2

Conclusion

The Library Management System was developed to provide a simple and efficient solution for managing the day-to-day activities of a library. The system helps to reduce manual work by maintaining book, student, and transaction records digitally. It provides an organized way to store and access important library information.

The system successfully supports important operations such as adding, updating, searching, issuing, and returning books. It also helps maintain accurate information about book availability and user records. By using a centralized database, the chances of data duplication, loss, and manual errors can be reduced. The project also provided practical knowledge of Java, Spring Boot, MySQL, HTML, CSS, and database connectivity. It helped in understanding how different technologies can be combined to develop a functional web-based application. The project also improved understanding of CRUD operations, database management, backend development, and user interface design.

Overall, the Library Management System provides a reliable, user-friendly, and organized platform for managing library activities. It saves time, reduces paperwork, and improves the efficiency of library operations. The system can be further enhanced in the future by adding features such as fine calculation, user authentication, email notifications, online book reservations, and detailed report generation.

References

1. Apache Maven Documentation — Maven documentation and build management. <https://maven.apache.org/guides/>
2. Baeldung – Learn Spring Boot — Tutorials and examples related to Spring Boot development. <https://www.baeldung.com/spring-boot>
3. Baeldung – Spring Framework Tutorial — Reference material for Spring, Spring MVC, JDBC, and database integration. <https://www.baeldung.com/spring-tutorial>
4. GitHub Documentation — Documentation for GitHub, repositories, version control, and software development workflows. <https://docs.github.com/en/get-started>
5. Java Documentation – Oracle — Official Java tutorials, API documentation, and developer guides. <https://docs.oracle.com/en/java/>
6. JDBC Tutorial – TutorialsPoint — Reference for Java Database Connectivity and relational database operations. <https://www.tutorialspoint.com/jdbc/index.htm>
7. MySQL 8.0 Reference Manual — Official documentation for MySQL database management and SQL. <https://dev.mysql.com/doc/refman/8.0/en/>
8. Spring Boot Reference Documentation — Official documentation for developing applications using Spring Boot. <https://docs.spring.io/spring-boot/reference/>
9. Spring Framework Documentation — Documentation and reference material for the Spring Framework and web application development. <https://docs.spring.io/spring-framework/reference/>
10. Thymeleaf Documentation — Official documentation for creating dynamic web pages using Thymeleaf. <https://www.thymeleaf.org/documentation.html>
11. TutorialsPoint – Java Tutorial — Learning resources and reference material for Java programming. <https://www.tutorialspoint.com/java/index.htm>
12. W3C – CSS Specifications — Reference for Cascading Style Sheets used for web page styling. <https://www.w3.org/Style/CSS/>
13. W3C – HTML Standard — Reference for HTML structure and web standards. <https://html.spec.whatwg.org/>
14. MDN Web Docs – HTML — Reference and tutorials for HTML, CSS, JavaScript, and web development. <https://developer.mozilla.org/en-US/docs/Web/HTML>
15. MDN Web Docs – CSS — Reference material for CSS concepts and web interface design. <https://developer.mozilla.org/en-US/docs/Web/CSS>
- 16.